

OS ORANGE PROBLEM-2

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Srn : PES1UG24CS352

Github link :

<https://github.com/SauravS9/PES1UG24CS352-pes-vcs>

1 A)

```
}
int hex_to_hash(const char *hex, ObjectID *id_out) {
    if (strlen(hex) < HASH_HEX_SIZE) return -1;
    for (int i = 0; i < HASH_SIZE; i++) {
        unsigned int byte;
        if (sscanf(hex + i * 2, "%2x", &byte) != 1) return -1;
        EOF return 0; += data_len; + 1, data_len); ze - data_offset; pe_out = OBJ_COMMIT; ze_
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:92:45: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=
]
    92 |         snprintf(tmp_path, sizeof(tmp_path), "%s.tmp", final_path);
        |         ~~~~~^~~~~~
In file included from /usr/include/stdio.h:980,
        from pes.h:8,
        from object.c:2:
In function 'snprintf',
    inlined from 'object_write' at object.c:92:5:
/usr/include/aarch64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size
512
    54 |     return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
        |           ~~~~~^~~~~~
    55 |                                     __glibc_objsize (__s), __fmt,
        |                                     ~~~~~^~~~~~
    56 |                                     __va_arg_pack ());
        |                                     ~~~~~^~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$
```

1B)

```
for (int i = 0; i < HASH_SIZE; i++) {
    unsigned int byte;
    if (sscanf(hex + i * 2, "%2x", &byte) != 1) return -1;
EOF return 0; += data_len; + 1, data_len); ze - data_offset; pe_out = OBJ_COMMIT; ze_
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:92:45: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=
]
92 |         snprintf(tmp_path, sizeof(tmp_path), "%s.tmp", final_path);
    |         ~~~~~^~~~~~
In file included from /usr/include/stdio.h:980,
    from pes.h:8,
    from object.c:2:
In function 'snprintf',
    inlined from 'object_write' at object.c:92:5:
/usr/include/aarch64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size
512
54 |     return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
    |           ^~~~~~
55 |                                     __glibc_objsize (__s), __fmt,
    |                                     ~~~~~~
56 |                                     __va_arg_pack ());
    |                                     ~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$
```

2A)

```

all: pes test_objects test_tree

clean:
    rm -f pes test_objects test_tree $(OBJS) test_objects.o test_tree.o
    rm -rf .pes

test: test-unit test-integration

test-unit: test_objects test_tree
    @echo "=== Running Phase 1 tests ==="
    ./test_objects
    @echo ""
    @echo "=== Running Phase 2 tests ==="
    ./test_tree

test-integration: pes
    @echo "=== Running integration tests ==="
    bash test_sequence.sh
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ cat >> tree.c << 'EOF'

/* Weak stub so test_tree links without index.o */
__attribute__((weak)) int index_load(Index *index) { (void)index; return -1; }
EOF
make test_tree && ./test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'write_tree_recursive':
tree.c:131:15: warning: implicit declaration of function 'object_write' [-Wimplicit-function-declaration]
  131 |     int ret = object_write(OBJ_TREE, data, len, id_out);
      |               ^
tree.c:120:13: warning: '__builtin___strncpy_chk' output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
  120 |         strncpy(te->name, dir_name, sizeof(te->name) - 1);
      |         ^
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$

```

2B)

```

rm -f pes test_objects test_tree $(OBJJS) test_objects.o test_tree.o
rm -rf .pes

test: test-unit test-integration

test-unit: test_objects test_tree
    @echo "=== Running Phase 1 tests ==="
    ./test_objects
    @echo ""
    @echo "=== Running Phase 2 tests ==="
    ./test_tree

test-integration: pes
    @echo "=== Running integration tests ==="
    bash test_sequence.sh
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ cat >> tree.c << 'EOF'

/* Weak stub so test_tree links without index.o */
__attribute__((weak)) int index_load(Index *index) { (void)index; return -1; }
EOF
make test_tree && ./test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'write_tree_recursive':
tree.c:131:15: warning: implicit declaration of function 'object_write' [-Wimplicit-function-declaration]
   131 |         int ret = object_write(OBJ_TREE, data, len, id_out);
       |               ^
tree.c:120:13: warning: '__builtin___strncpy_chk' output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
   120 |         strncpy(te->name, dir_name, sizeof(te->name) - 1);
       |         ^
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ xxd .pes/objects/$(ls .pes/objects/ | head -1)/$(ls .pes/objects/$(ls .pes/objects/ | head -1)/
| head -1) | head -20
xxd: Is a directory
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$

```

3A)

```

EOFnt("Done")ew_content)) as f: new_save + content[end:]
Done
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ make pes
rm -rf .pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function 'index_add':
index.c:167:5: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
   167 |     fread(data, 1, size, f);
       |     ^~~~~~
index.c: At top level:
index.c:98:12: warning: 'compare_index_entries' defined but not used [-Wunused-function]
    98 | static int compare_index_entries(const void *a, const void *b) {
       |            ^~~~~~
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
staged:    file1.txt
staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
untracked: object.c
untracked: index.h
untracked: pes.h
untracked: .gitignore
untracked: tree.h
untracked: test_objects
untracked: tree.c
untracked: commit.h
untracked: commit.c
untracked: README.md
untracked: test_tree
untracked: test_objects.c

```

3B)

```
index.c:167:5: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
167 |     fread(data, 1, size, f);
    |     ^
index.c: At top level:
index.c:98:12: warning: 'compare_index_entries' defined but not used [-Wunused-function]
98 | static int compare_index_entries(const void *a, const void *b) {
    |
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
staged:    file1.txt
staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
untracked: object.c
untracked: index.h
untracked: pes.h
untracked: .gitignore
untracked: tree.h
untracked: test_objects
untracked: tree.c
untracked: commit.h
untracked: commit.c
untracked: README.md
untracked: test_tree
untracked: test_objects.c
untracked: index.c
untracked: test_sequence.sh
untracked: Makefile
untracked: test_tree.c
untracked: pes.c

rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776876802 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776876802 6 file2.txt
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$
```

4A)

```

echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
[main 01ac219] Initial commit
Committed: 01ac2199371d... Initial commit
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
./pes log
[main f83f755] Add world
Committed: f83f75523eed... Add world
[main 10d5b08] Add farewell
Committed: 10d5b08c79a3... Add farewell
commit 10d5b08c79a31f5f806c1ab0194eb28b21a01395606442dfa4dad6f10235213b
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877429

    Add farewell!

commit f83f75523eed405405fd1d90b37969ac8a524b83152a76f1cfb9d6d1fda02042
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877429

    Add worldell!

commit 01ac2199371df0a032fa77035bf281a1ef767b20d37fd3a4971757e3e205f6d5
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877393

    Initial committial cA

rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$

```

4B)

```

./pes add bye.txt
./pes commit -m "Add farewell"
./pes log
[main f83f755] Add world
Committed: f83f75523eed... Add world
[main 10d5b08] Add farewell
Committed: 10d5b08c79a3... Add farewell
commit 10d5b08c79a31f5f806c1ab0194eb28b21a01395606442dfa4dad6f10235213b
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877429

    Add farewell!

commit f83f75523eed405405fd1d90b37969ac8a524b83152a76f1cfb9d6d1fda02042
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877429

    Add worldell!

commit 01ac2199371df0a032fa77035bf281a1ef767b20d37fd3a4971757e3e205f6d5
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877393

    Initial committial cA

rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/01/ac2199371df0a032fa77035bf281a1ef767b20d37fd3a4971757e3e205f6d5
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/10/d5b08c79a31f5f806c1ab0194eb28b21a01395606442dfa4dad6f10235213b
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375ffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f8/3f75523eed405405fd1d90b37969ac8a524b83152a76f1cfb9d6d1fda02042
.pes/refs/heads/main
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$

```

4C)

```

Committed: f83f75523eed... Add world
[main 10d5b08] Add farewell
Committed: 10d5b08c79a3... Add farewell
commit 10d5b08c79a31f5f806c1ab0194eb28b21a01395606442dfa4dad6f10235213b
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877429

    Add farewell!

commit f83f75523eed405405fd1d90b37969ac8a524b83152a76f1cfb9d6d1fda02042
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877429

    Add worldell!

commit 01ac2199371df0a032fa77035bf281a1ef767b20d37fd3a4971757e3e205f6d5
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877393

    Initial committial cA

rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/01/ac2199371df0a032fa77035bf281a1ef767b20d37fd3a4971757e3e205f6d5
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/10/d5b08c79a31f5f806c1ab0194eb28b21a01395606442dfa4dad6f10235213b
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375ffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f8/3f75523eed405405fd1d90b37969ac8a524b83152a76f1cfb9d6d1fda02042
.pes/refs/heads/main
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
10d5b08c79a31f5f806c1ab0194eb28b21a01395606442dfa4dad6f10235213b
ref: refs/heads/main
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$

```

Final screenshot 1 :

```
ref: refs/heads/main
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:   file.txt
  staged:   hello.txt

Unstaged changes:
(nothing to show)

Untracked files:
(nothing to show)

--- First Commit ---
[main f3c3652] Initial commit
Committed: f3c36526ac7b... Initial commit

Log after first commit:
commit f3c36526ac7b63d9f5256cef9550fd1293d77adbfceacfbe0e86082cbbc8396c
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date:   1776877535

    Initial commit

--- Second Commit ---
[main 6e4dd9a] Update file.txt
Committed: 6e4dd9a64f13... Update file.txt
```

Final screenshot 2 :

```
Date: 1776877535

Add farewell!

commit 6e4dd9a645133c08fe70139d5769d88df159b738b837804164fb219b4f6eddeb
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877535

Update file.txt

commit f3c36526ac7b63d9f5256cef9550fd1293d77adbfceacfbe0e86082cbbc8396c
Author: R SAURAV SRRINIVAS <PES1UG24CS352>
Date: 1776877535

Initial committial cA

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
263583cdf7e2ab95eaacad36e642b8fb53e3c6567f40647e8332e19d0ac8133

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/26/3583cdf7e2ab95eaacad36e642b8fb53e3c6567f40647e8332e19d0ac8133
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/4dd9a645133c08fe70139d5769d88df159b738b837804164fb219b4f6eddeb
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08da0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaea3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/f3/c36526ac7b63d9f5256cef9550fd1293d77adbfceacfbe0e86082cbbc8396c

I

=== All integration tests completed ===
rsaurav@PES1UG24CS352:~/PES1UG24CS352-pes-vcs$
```